# Data Warehousing and Data Mining – Practical Lab Manual

## Practical 1: Visualization Features in WEKA

Objective:

Explore the visualization features of WEKA for data analysis.

Tasks:

- Load sample datasets (e.g., Iris).

- Use the Visualize tab to plot attributes.

- Identify patterns, correlation, and outliers.

Expected Outcome:

Ability to interpret visual patterns in datasets.

## Practical 2: Data Preprocessing & Association Rule Mining

Objective:

Perform data preprocessing and generate association rules.

Tasks:

- Clean and preprocess dataset (handling missing values, normalization).

- Apply Apriori algorithm.

- Generate association rules using support & confidence.

Expected Outcome:

Understanding of preprocessing and association rule generation.

## Practical 3: Classification on Datasets

Objective:

Apply classification algorithms using WEKA.

Tasks:

- Use algorithms (J48, Naive Bayes, SVM).

- Train/test using percentage split.

- Compare accuracy metrics.

Expected Outcome:

Ability to classify datasets using multiple algorithms.

## Practical 4: Clustering on Datasets

Objective:

Perform clustering techniques.

Tasks:

- Use K-Means, Hierarchical clustering.

- Analyze clusters using visualization.

- Interpret cluster grouping patterns.

Expected Outcome:

Ability to perform unsupervised learning tasks.

## Practical 5: Programs Using German Credit Data

Objective:

Apply ML techniques to German Credit dataset.

Tasks:

- Load dataset into WEKA.

- Perform preprocessing.

- Classify and evaluate results.

Expected Outcome:

Hands-on experience with real-world dataset.

## Practical 6: Decision Tree Using Cross Validation

Objective:

Understand cross-validation in decision tree training.

Tasks:

- Train decision tree using 10-fold cross-validation.

- Compare with training/testing split.

- Discuss accuracy differences.

Expected Outcome:

Knowledge of model validation techniques.

## Practical 7: Attribute Bias Analysis

Objective:

Identify bias in dataset attributes.

Tasks:

- Test if data has bias against 'foreign workers' or 'personal-status'.

- Remove attributes and test performance.

- Document impact.

Expected Outcome:

Understanding of attribute relevance and bias.

## Practical 8: Attribute Combination Testing

Objective:

Test model performance with reduced attribute combinations.

Tasks:

- Select subsets of attributes.

- Compare accuracy and performance.

- Discuss results.

Expected Outcome:

Understanding dimensionality and feature selection.

## Practical 9: Decision Tree vs Cross-validation

Objective:

Compare decision tree training methods.

Tasks:

- Train model regularly.

- Train model with cross-validation.

- Compare accuracy, confusion matrix, model complexity.

Expected Outcome:

Clear understanding of model reliability.

## Practical 10: Decision Tree Complexity vs Bias

Objective:

Study relationship between model complexity and bias.

Tasks:

- Train trees with different pruning levels.

- Compare bias-variance tradeoff.

- Document learning outcomes.

Expected Outcome:

Knowledge of bias-variance relationship in ML.